

How Searchlight Learns

A plain-English guide to the "smart detector" in Searchlight

The one-line version

Searchlight already guesses, for every site it checks, whether your target **really has an account there**. You can make those guesses *better over time* by telling it when it got one right or wrong. That's the whole feature. It learns **only from you, only on your machine**, and it stays off until *your* version is provably better than the one it ships with.

1. What the detector is doing in the first place

When you run a username sweep, Searchlight visits each site and has to decide one of three things:

-  **Found** — looks like a real account.
-  **Maybe** — it's not sure. The page was ambiguous.
-  **Not found** — looks like there's no account.

Most sites are easy (the site flat-out says "user not found"). The hard ones land in **Maybe** — and *those* are where your help is worth the most.

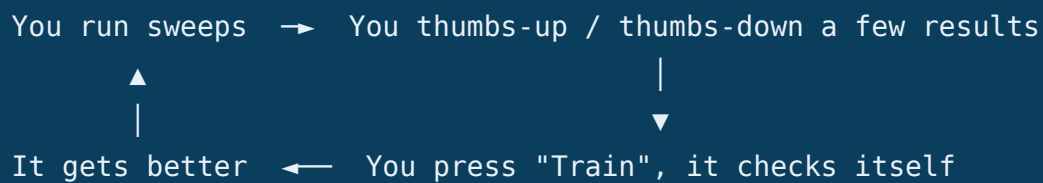
You don't have to do anything for this part. It happens automatically on every sweep. The learning feature below is just about making the **Maybe** pile smaller and the / calls more trustworthy.

2. The big idea: it learns from *you*, not from the internet

Searchlight ships with one **generic** detector — a starting point trained on a small public sample. It's decent, not brilliant.

The catch with any generic detector is that *your* cases aren't average. You work specific sites, specific kinds of targets. So instead of trying to be clever for everyone, Searchlight lets **your own corrections** retrain a detector that's tuned to **your** work.

Here's the loop, and it's short:



Nothing leaves your computer. There is no shared model, no upload, no "anonymous data to improve the product." Your corrections train a file that lives only in your own app data. (More on privacy in section 6.)

3. Why it starts switched OFF (and that's on purpose)

When you first install, the smart detector is **off**. The Settings screen even labels it *"experimental; bundled model pending retrain."* That's honest, not a bug.

The reason: a generic model that hasn't learned your cases yet is, at best, a tie with the plain detector. Turning it on before it has earned its place would just add risk for no gain. So Searchlight makes you **earn the upgrade** — it won't even let you *check* whether the smart detector is better until you've given it enough corrections to judge fairly.

Think of it like this: **off by default = "prove it first."**

4. How to teach it — step by step

You teach it by **labeling** results: a 👍 or 👎 on results Searchlight wasn't sure about.

Where the buttons are

You can label in **two places** — use whichever fits your flow:

1. **Right on the sweep results** — every ✅ Found and ? Maybe row has a small 👍 / 👎 next to it (you'll only see them when a case is open). Tap one and a ✓ replaces it. Done.
2. **The "Learning" tab** (inside Searchlight, next to Sweep) — this is the calmer way. It hands you a **short, prioritized list** — at most **10 of the most useful "Maybe" results at a time** — so you're never staring at a wall. Label them, they drop off the list, next chunk appears.

What the two buttons mean

- 👍 **Real** — "yes, this account is genuinely theirs."
- 👎 **Not real** — "no, that's a false hit / not them."

That's the entire skill. You're not scoring anything. You're just confirming or correcting.

The one number to watch: 80

Under the action button there's a progress bar toward **80 labels**. Eighty is the point where Searchlight has *enough* of your corrections to test itself honestly. Until then it'll just say:

"Keep labeling — 34/80 until your model can be checked."

You don't have to do all 80 in one sitting. Label a handful after each sweep; it remembers. The bar fills up as you go.

5. The "Learning" tab, button by button

The Learning tab always shows you **one** thing to do next and **one** plain sentence about where you stand — never a pile of statistics. Here's every state you might see, and what it means:

What the button says	What it means	What to do
Label results to teach the detector	You're still building up corrections. <i>"Keep labeling — N/80 until your model can be checked."</i>	Keep thumbs-up/down-ing results. Watch the bar climb to 80.
Train now	You hit 80+. <i>"You have enough labels — train to check if your model beats the built-in detector."</i>	Press it. It builds your detector and tests it (takes a moment).
Enable — beats the built-in detector	Your detector passed the test. <i>"Your model now beats the built-in detector on your cases."</i>	Press it to switch your smarter detector ON.
Train again	It trained but didn't beat the default yet. <i>"Not yet — your model doesn't beat the built-in detector. Label more, then retrain."</i>	Label more results, then press it again. No harm done.
Retrain	Your smart detector is already ON. <i>"ML is on — beating the built-in detector on your cases."</i>	Optional. As you label more, retrain occasionally to keep it sharp.

The safety net: if you retrain and the new version *isn't* actually better, Searchlight keeps your old working one. A bad retrain can never make your detector worse. So pressing "Retrain" is always safe.

6. The privacy promise (why this is safe to use)

This feature was built to *not* leak anything:

- **Local only.** Your corrections, the captured page-features, and your trained detector all live in your own app data folder. There is **no central model** and **no network connection** in the learning part at all. The only time Searchlight touches the network is the actual username-checking on sites — which is the same Tor-gated switch you already control.
- **Encrypted at rest — when you're logged in.** If you use the app with a vault/login, all of it (your labels, your model) is encrypted on disk. *Note:* if you run the app with **no login set up**, those files sit in plain text — so if this is sensitive casework, use a login.

- **You can't accidentally poison it.** The buttons only send "this result, thumbs-up/down" — the actual page data it learns from is captured by the app itself during the sweep, not typed in. So a stray click is just a stray click, easily corrected by labeling it the other way.
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7. For the curious: the Settings switches (optional)

You never *need* to touch these — the Learning tab handles everything. But if you open **Settings** → **Searchlight** you'll find:

- **Deep scan** (on by default) — lets Searchlight take a second, closer look at ambiguous sites. Leaving it on gives better answers; turning it off makes sweeps faster but leaves more results in "Maybe."
 - **Use ML model** — the same on/off switch as the Learning tab's Enable button. Marked *experimental* for the reason in section 3.
 - **Found / Maybe thresholds** — advanced dials for *how confident* a result must be to count as Found or Maybe. The placeholders show the model's own defaults. **Leave these blank unless you know exactly why you're changing them** — the detector's own calibrated values are almost always better than a hand-picked number.
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Quick reference card

To make Searchlight smarter:

1. Run sweeps like normal.
2. After each, open the **Learning** tab and 👍/👎 a few **Maybe** results.
3. Watch the bar climb to **80**.
4. Press **Train now**.
5. If it says "**beats the built-in detector**", press **Enable**.
6. Every so often, **Retrain** to keep it sharp.

It learns only from you, only on your machine. You can't break it — a worse retrain is always thrown away.

Ghost Intel 98 — Searchlight. This guide describes the adaptive-learning feature as shipped in v3.24.x.